

WEEK 1-2: Vector Spaces, Linear Maps, Duality, Operators

Field F: a collection of scalars with addition and multiplication that behave normally

- Contains a zero element ($0 + x = x$)
- Contains a one element ($1 * x = x$)
- Each non-zero element has a additive and multiplicative inverse

Vector space V: a set of vectors over field F that is

- Closed under vector addition
- Closed under scalar multiplication
- Additive commutativity (order doesn't matter)
- Additive associativity (grouping doesn't matter)
- Additive identity (contains zero vector) & inverse
- Multiplicative identity & associativity
- Distributive property for scalar into sum of vectors & vector into sum of scalars

Subspace U: a subset of a vector space V that is a vector space itself.

- Contains zero vector
- Closed under addition
- Closed under scalar multiplication

Linear Combination: sum of vectors scaled by elements of the field $c_1v_1 + \dots + c_nv_n$

Span: the set of all possible linear combinations of a list of vectors (a list of vectors spans V means any vector in V can be built out of linear combinations of the vectors, or any v in V can be written as $v = a_1v_1 + \dots$)

Linear Independence: a list of vectors ($v_1, \dots, v_n$) is linearly independent if $c_1v_1 + \dots + c_nv_n = 0$ forces $c_1 = \dots = c_n = 0$

Basis: a list of vectors that is linearly independent and spans the entire vector space. Any LI list with whose length equals dimension of vector space automatically spans the space and forms a basis

Span/LI Lemma: the length of any linearly independent list of vectors cannot exceed $\leq$ the length of any spanning list of vectors.

Dimension: the number of elements contained in a basis

Direct sum ($U \oplus W$): V is a direct sum of U and W if every v in V can be written uniquely as $v = u + w$, where u in U and w in W. In other words, $U + W = V$ and $U \cap W = \{0\}$

Linear map: a function/transformation $V \rightarrow W$ if for all u and v in V and all scalars c in F

- $f(u+v) = f(u) + f(v)$
- $f(cu) = cf(u)$

Null Space of linear map $T \in L(V, W) = \{v \in V : T(v) = 0\}$; Null T is a subspace of V (domain)

Range of arbitrary function $T \in L(V, W)$ is a set of all outputs w such that v is in $V = \{w = T(v) : v \in V\}$; Range T is a subspace of W

A function $T: V \rightarrow W$ is **injective** if every element in range T has a unique vector that maps to it from domain. Let $T \in L(V, W)$, then T is injective if and only if $\text{null } T = \{0\}$.

A function $T: V \rightarrow W$ is **surjective** if $\text{range } T = W$

A in $L(V)$ is **invertible** if there exists A^{-1} such that $AA^{-1} = I$ and $A^{-1}A = I$

If V is finite dimensional (or $\dim(V) = \dim(W)$), then T invertible $\Leftrightarrow T$ injective $\Leftrightarrow T$ surjective.

Rank-nullity theorem (V finite dim): $\dim(V) = \dim(\text{null } T) + \dim(\text{range } T)$

- So if $\dim(V) > \dim(W)$, T can never be injective; if $\dim(V) < \dim(W)$, T can never be surjective
- "Fundamental theorem of linear maps" — the real workhorse for dimension-counting arguments

Product space ($V_1 \times V_2 \times \dots \times V_n$): the set of all ordered tuples $(v_1, v_2, \dots, v_n)$ where $v_i \in V_i$

- Add tuples component-by-component, scale tuples by every component
- $\dim(V_1 \times \dots \times V_n) = \dim(V_1) + \dots + \dim(V_n)$

Gamma map: $\Gamma(u_1, \dots, u_n) = u_1 + \dots + u_n$, for subspaces $U_1, \dots, U_n$ that live inside the same space V

- $U_1 + \dots + U_n$ is a direct sum $\Leftrightarrow \Gamma$ is injective $\Leftrightarrow \dim(U_1 + \dots + U_n) = \dim(U_1) + \dots + \dim(U_n)$

- Checking that every pair intersects at only $\{0\}$ is NOT enough to prove a direct sum for 3+ subspaces

Translate (coset): take a subspace U and shift it by adding a fixed vector v to every vector in it: $v + U = \{v+u : u \text{ in } U\}$. Same shape as U , just moved so it passes through v instead of through 0 .

- **Coset equality:** $v + U$ and $w + U$ are literally the same set exactly when $v - w$ is in U

Quotient space V/U : take all the cosets of U and treat each one as a single point. You can add and scale cosets by adding/scaling their anchor vectors.

- $\dim(V/U) = \dim(V) - \dim(U)$

Quotient map (π): $\pi(v) = v + U$. Always linear, always surjective onto V/U , and $\text{null}(\pi) = U$

Induced map ($T_{\sim}$): given T in $L(V,W)$, define $T_{\sim}$ on $V/(\text{null } T)$ by $T_{\sim}(v + \text{null } T) = Tv$

- $T_{\sim}$ is always injective, and $\text{range}(T_{\sim}) = \text{range}(T)$
- So $V/(\text{null } T)$ is isomorphic to $\text{range}(T)$ — always true, no matter what T is

Linear functional: a linear map that spits out a single number instead of a vector (its output space is F).

Dual space (V'): the set of all linear functionals on V .

- $\dim(V') = \dim(V)$

Dual basis: given a basis $v_1, \dots, v_n$ of V , the dual basis $\Phi_1, \dots, \Phi_n$ is the set of functionals where Φ_i reads off exactly the i -th coefficient of any vector.

- $\Phi_i(v_j) = 1$ if $i = j$, and 0 if i is not j
- Shortcut for basis $1, x, x^2, \dots, x^n$ of a polynomial space: $\Phi_k(p) = (k\text{-th derivative of } p, \text{ evaluated at } 0) / k!$
- If the basis is centered somewhere else (like $1, (x-5), (x-5)^2, \dots$), evaluate the derivative at that point instead of 0

Dual map (T'): for T in $L(V,W)$, the dual map T' goes the opposite direction: T' is in $L(W', V')$

- Definition: $T'(\Phi)(v) = \Phi(Tv)$

Annihilator (U_0): the set of every functional in V' that gives 0 on every single vector in U .

- $\dim(U_0) = \dim(V) - \dim(U)$
- $U_0 = \{0\}$ exactly when $U = V$
- $U_0 = V'$ exactly when $U = \{0\}$

Null space and range of T' :

- $\text{null}(T') = (\text{range } T)_0$
- $\text{range}(T') = (\text{null } T)_0$
- T surjective $\Leftrightarrow T'$ injective
- T injective $\Leftrightarrow T'$ surjective

Zero-counting fact: a polynomial of degree m (that isn't the zero polynomial) has at most $\leq m$ zeros (roots).

Factor theorem: if $p(\lambda) = 0$, you can pull out a factor: $p(z) = (z - \lambda) * q(z)$, where q has one degree less than p .

Division algorithm: for any polynomials p and q (q not zero), you can always write $p = s*q + r$, where r has a smaller degree than q . This is just long division, but with polynomials instead of numbers.

Fundamental theorem of algebra: every polynomial with complex coefficients (degree 1 or higher) has at least one root somewhere in the complex numbers. Because of this, every complex polynomial can be split completely into pieces that look like $(z - \lambda)$.

Real polynomial factoring: over the real numbers, a polynomial splits into $(z - \lambda)$ pieces for real roots, plus leftover quadratic pieces (like $z^2 + bz + c$, with no real root) for pairs of complex roots that come as conjugates.

Operator: a linear map $T: V \rightarrow V$, meaning the input and output live in the exact same space.

$p(T)$: plug the operator T into a polynomial p . If $p(z) = a_0 + a_1z + \dots + a_nz^n$, then $p(T) = a_0I + a_1T + a_2T^2 + \dots + a_nT^n$

- Two different polynomials plugged into the same T always commute: $p(T)*q(T) = q(T)*p(T)$

Invariant subspace (U): a subspace where T never sends you outside of U . In symbols: Tu is in U for every u in U .

- Always automatically invariant, no matter what T is: $\{0\}$, V itself, $\text{null}(T)$, and $\text{range}(T)$
- Bigger fact: $\text{null}(p(T))$ and $\text{range}(p(T))$ are invariant under T itself, for any polynomial p

Eigenvalue (λ): a number where $Tv = \lambda v$ for some nonzero vector v .

- Same as saying: $T - \lambda I$ is not injective (and therefore not invertible either)
- Eigenvectors that come from different (distinct) eigenvalues are always linearly independent
- An operator has at most $\dim(V)$ different eigenvalues total

Eigenvector: the nonzero vector v itself that satisfies $Tv = \lambda v$.

Eigenspace $E(\lambda, T)$: all the eigenvectors for one particular λ , plus the zero vector. $E(\lambda, T) = \text{null}(T - \lambda I)$.

When are eigenvalues guaranteed to exist?

- Always guaranteed on a nonzero, finite dimensional complex space
- NOT guaranteed on a real space — example: rotating the plane by 90 degrees has no real eigenvalue at all
- Exception: if the real space has odd dimension, at least one real eigenvalue is guaranteed

Minimal polynomial: the smallest-degree monic (leading coefficient = 1) polynomial p such that $p(T) = 0$. Every operator has exactly one of these.

- Its degree is always at most $\dim(V)$
- Any polynomial q satisfies $q(T) = 0$ exactly when q is a multiple of the minimal polynomial
- The zeros (roots) of the minimal polynomial are exactly the eigenvalues of T . Each distinct eigenvalue shows up only once in the factored form, even if it "feels" like it should repeat
- T is invertible exactly when the constant term (the number with no z attached) of the minimal polynomial is not zero

Upper triangular matrix (for T , w.r.t. basis $v_1, \dots, v_n$): every entry below the diagonal is 0. Three equivalent ways to say this is true:

- $\text{span}(v_1, \dots, v_k)$ is invariant under T , for every single k from 1 to n
- Tv_k lands inside $\text{span}(v_1, \dots, v_k)$, for every k
- The matrix literally has zeros below the diagonal

If the matrix is upper triangular: the numbers sitting on the diagonal are exactly the eigenvalues of T (repeats are allowed).

Careful: this does not mean every basis vector is an eigenvector — only the very first one, v_1 , is guaranteed to actually be one.

When can you triangularize an operator?

- Over the complex numbers: always possible, for any nonzero finite dimensional operator
- Over the real numbers or complex numbers (general test): possible exactly when the minimal polynomial factors completely into linear $(z - \lambda)$ pieces over that field

Diagonalizable: T 's matrix comes out completely diagonal (all zeros except the diagonal) with respect to some basis. Four ways to say the same thing:

- V has a basis made entirely of eigenvectors of T
- V splits into a direct sum of all of T 's eigenspaces: $V = E(\lambda_1, T) (+) E(\lambda_2, T) (+) \dots (+) E(\lambda_m, T)$
- $\dim(V)$ equals the eigenspace dimensions all added together
- T 's matrix is diagonal in some basis

Minimal polynomial of T factors as $p(x) = (x - \lambda_1) \dots (x - \lambda_m)$ such that λ_i 's are distinct

Easy shortcut: if T has $\dim(V)$ different eigenvalues (the maximum possible amount), it is automatically diagonalizable. (But this is not required — you can be diagonalizable with fewer distinct eigenvalues too, as long as the eigenspaces are big enough to fill up the space.)

Best test when eigenvalues are ugly or unknown: T is diagonalizable exactly when its minimal polynomial has no repeated roots.

Always true, whether or not diagonalizable: adding up the dimensions of every eigenspace never exceeds $\dim(V)$. It equals $\dim(V)$ exactly when T is diagonalizable.

If T is diagonalizable and U is invariant under T , then T restricted to just U is also diagonalizable.

Gershgorin disk theorem: every eigenvalue of a matrix sits inside a disk centered at one of the diagonal entries A_{ii} , with radius equal to the sum of the absolute values of the rest of that row. Useful for proving invertibility without finding exact eigenvalues.

Similar operators: R and T are called similar if $R = S^{-1} * T * S$ for some invertible S . This means R and T are really "the same" transformation, just written in two different bases.

Commuting operators: S and T commute if $ST = TS$ (order doesn't matter).

Key calculation: if S and T commute, and v is an eigenvector of T with eigenvalue λ , then:

- $T(Sv) = S(Tv) = S(\lambda v) = \lambda(Sv)$
- This means Sv is also an eigenvector of T with the same eigenvalue λ (or it's zero). In other words, S never pushes T 's eigenvectors out of T 's eigenspaces.

Bigger facts that follow from this:

- S and T are simultaneously diagonalizable (share one basis that diagonalizes both) exactly when they commute
- Any two commuting operators on a nonzero, finite dimensional complex space share at least one common eigenvector (not necessarily the same eigenvalue for it though)
- Two commuting operators on a complex space can always be triangularized using the same shared basis
- If S and T commute: every eigenvalue of S+T is (an eigenvalue of S) plus (an eigenvalue of T), and every eigenvalue of ST is (an eigenvalue of S) times (an eigenvalue of T). This completely breaks if S and T do not commute.

WEEK 3: Inner Product Spaces, Adjoints, Spectral Theorems

Inner product $\langle \cdot, \cdot \rangle$ on V (over $F = \mathbb{R}$ or $\mathbb{C}$): a function $V \times V \rightarrow F$ that is

- Linear in the 1st slot ($\langle au+bv, w \rangle = a\langle u, w \rangle + b\langle v, w \rangle$)
- Conjugate symmetric: $\langle v, w \rangle = \text{conj}(\langle w, v \rangle)$ (just = for $\mathbb{R}$)
- Positive definite: $\langle v, v \rangle \geq 0$, and = 0 only when $v = 0$

Norm: $\|v\| = \sqrt{\langle v, v \rangle}$. **Cauchy-Schwarz:** $|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$, equality iff u, v parallel. **Triangle inequality:** $\|u+v\| \leq \|u\| + \|v\|$.

Orthogonal complement ($U^\perp$): $\{v \in V : \langle v, u \rangle = 0 \text{ for all } u \in U\}$

- $V = U \oplus U^\perp$ (always, for U a subspace of finite-dim V) — orthogonal projection P_U exists and is unique
- $(U^\perp)^\perp = U$; $\dim(U^\perp) = \dim(V) - \dim(U)$

Adjoint (T^*): for $T \in L(V, W)$, T^* satisfies $\langle Tv, w \rangle = \langle v, T^*w \rangle$ for all v, w

- Algebraic rules: $(S+T)^* = S^*+T^*$, $(\lambda T)^* = \lambda T^*$, $(ST)^* = T^*S^*$, $(T^{-1})^* = (T^*)^{-1}$, $(T^*)^* = T$
- Matrix fact (standard inner product $\langle x, y \rangle = y^*x$): $T^* = A^*$ (conjugate transpose)
- Weighted inner product ($\langle x, y \rangle = y^*Wx$, W Hermitian positive-definite): $T^* = W^{-1}A^*W$
- To find T^* by hand: write LHS = $\langle Ax, y \rangle$, RHS = $\langle x, By \rangle$ using the given inner product, expand both in matrix form, match them, solve for $B = T^*$

Riesz representation: every linear functional $\phi: V \rightarrow F$ can be written $\phi(p) = \langle p, q \rangle$ for a unique $q \in V$

- To find q: build an orthonormal basis $e_1, \dots, e_n$ (Gram-Schmidt if needed), compute $\phi(e_i)$ for each i, then $q = \sum \phi(e_i) \cdot e_i$

Normal operator: $T^*T = TT^*$ (self-adjoint is a special case)

- T normal $\Leftrightarrow \|Tv\| = \|T^*v\|$ for all v
- Normal $\Rightarrow$ eigenvectors from distinct eigenvalues are orthogonal (not just LI)
- Normal $\Rightarrow \text{null}(T) = \text{null}(T^*)$ and $\text{range}(T) = \text{range}(T^*)^\perp$ -- wait, more precisely $\text{range}(T)^\perp = \text{null}(T^*)$ always holds (any T), and = $\text{null}(T)$ when normal

Complex Spectral Theorem (V complex, finite dim): T is normal $\Leftrightarrow$ V has an orthonormal basis of eigenvectors of T (T diagonal in an orthonormal basis)

Real Spectral Theorem (V real, finite dim): T is self-adjoint $\Leftrightarrow$ V has an orthonormal basis of eigenvectors of T, all eigenvalues real

- This is the "always diagonalizable" shortcut for self-adjoint operators — don't need to check min poly
- Self-adjoint $\nRightarrow$ PSD (eigenvalues can be negative); PSD needs self-adjoint + eigenvalues ≥ 0

Self-adjoint: $T^* = T \Leftrightarrow \langle Tv, v \rangle \in \mathbb{R}$ for all v

Positive semidefinite (PSD): all equivalent:

- $T^* = T$ and $\langle Tv, v \rangle \geq 0$ for all v
- $T = T^*$ and all eigenvalues ≥ 0
- Diagonal (entries ≥ 0) in some orthonormal basis
- T has a unique PSD square root
- T has some self-adjoint square root
- $T = R^*R$ for some R

Square root ($\sqrt{T}$): $S^2 = T$. If T is PSD, the PSD square root is unique.

- Diagonalize $T = PDP^{-1}$ with orthonormal eigenbasis (so $P^{-1} = P^*$), confirm all eigenvalues $\lambda_i \geq 0$, form $\sqrt{D} = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})$, then $\sqrt{T} = P\sqrt{D}P^{-1}$

Rank-one perturbation operators: $T(v) = av + b \cdot \text{proj}_u(v)$, where $\|u\| = 1$

- Eigenvalue $a+b$ on $\text{span}(u)$; eigenvalue a (multiplicity $n-1$) on $u^\perp$
- Automatically self-adjoint for real a, b , since proj_u is self-adjoint
- PSD exactly when $a \geq 0$ and $a+b \geq 0$
- $\sqrt{T}(v) = \sqrt{a} \cdot v + (\sqrt{a+b} - \sqrt{a}) \cdot \text{proj}_u(v)$
- $T^{-1}(v) = (1/a)v + (1/(a+b) - 1/a) \cdot \text{proj}_u(v)$, valid when $a \neq 0, a+b \neq 0$

Commuting fact: if $ST = TS$, then $S\sqrt{T} = \sqrt{T}S$ (proof: spectral theorem gives $T = \sum \lambda_i \cdot P_i$ where P_i projects onto eigenspace E_i ; S preserves each E_i since $T(Sv) = S(Tv) = \lambda_i(Sv)$, so S commutes with each P_i , hence with $\sqrt{T} = \sum \sqrt{\lambda_i} \cdot P_i$)

Orthonormal: every column vector in a matrix has norm = 1 and is orthogonal to every other column vector (dot product = 0)

Isometry: a linear map $S: V \rightarrow W$ is an isometry if $\|Sv\| = \|v\|$ (with orthonormal bases $(e_1, \dots, e_n)$ for V and $(f_1, \dots, f_n)$ for W). These are all equivalent:

- S is an isometry ($\|Sv\| = \|v\|$)
- $S^*S = I$
- $\langle Su, Sv \rangle = \langle u, v \rangle$ for all u, v in V
- $Se_1, \dots, Se_n$ are orthonormal (not necessarily an orthonormal basis of W)
- Columns of $M(S)$ are orthonormal
- Eigenvalues have norm = 1

A matrix Q ($m \times n$) is an isometry iff its columns are orthonormal (only possible if Q is tall-skinny or square)

Unitary operator: a special type of isometry $S \in L(V)$. These are all equivalent:

- S is unitary
- $SS^* = S^*S = I$
- $S^{-1} = S^*$
- $Se_1, \dots, Se_n$ is an orthonormal basis of V
- Rows of $M(S)$ form an orthonormal basis
- S^* is unitary

QR factorization: any matrix A with linearly independent columns can be written $A = QR$, where Q has orthonormal columns and R is upper triangular.

Existence of the singular value decomposition (for $T \in L(V, W)$, any linear map): These are all equivalent:

- T^*T is positive semidefinite (EQUIVALENT TO A LOT OF OTHER STUFF)
- $\text{null}(T^*T) = \text{null}(T)$
- $\text{range}(T^*T) = \text{range}(T^*)$
- $\dim(\text{range } T) = \dim(\text{range } T^*) = \dim(\text{range } T^*T)$

Singular values: the positive square roots of the eigenvalues of T^*T , counted with multiplicity.

Facts about singular values:

- The number of strictly positive singular values equals $\dim(\text{range } T)$
- T is injective $\Leftrightarrow 0$ is not a singular value
- T is surjective $\Leftrightarrow$ number of positive singular values = $\dim W$
- T is an isometry $\Leftrightarrow$ every singular value of T equals 1

Singular Value Decomposition (SVD): suppose $T \in L(V, W)$ with positive singular values $\sigma_1, \dots, \sigma_m$. Then there exist orthonormal lists $(e_1, \dots, e_m)$ for V and $(f_1, \dots, f_m)$ for W ($f = T(e)/\sigma$) such that $Tv = \sigma_1 \langle v, e_1 \rangle f_1 + \dots + \sigma_m \langle v, e_m \rangle f_m$. Matrix form: $A = U \Sigma V^T$

WEEK 4: Structure Theory — Generalized Eigenvectors, Nilpotents, Jordan Form

Krylov subspace (order k): for $T \in L(V)$ and $v \in V$, then $K_k(v) = \text{span}(v, Tv, T^2v, \dots, T^{k-1}v)$.

- If v is an eigenvector of T , every Krylov subspace of v is just $\text{span}(v)$, since $\text{span}(Tv) = \text{span}(v)$ already
- $K_1(v) \subset K_2(v) \subset \dots \subset K_k(v)$ is a nested sequence of subspaces
- If $K_k = K_{k+1}$, then $K_k = K_i$ for all $i > k$ (proof by algebra + induction)
- For any T and v , there is some r such that $\dim(K_k(v)) = k$ for $k \leq r$, and $\dim(K_k(v)) = r$ for $k > r$
- If the full Krylov subspace has dimension r , then $K_r(v)$ is T -invariant (a subspace that stays inside itself under T)

Minimal polynomial of T with respect to v: the unique monic polynomial p_v of smallest degree such that $p_v(T)v = 0$.

- Any polynomial q with $q(T)v = 0$ must be a multiple of p_v

Theorem: let $r = \dim$ of the full Krylov subspace of v . Then $r = \deg(p_v)$.

- $v, Tv, \dots, T^{r-1}v$ is linearly independent, while $v, Tv, \dots, T^r v$ is linearly dependent
- Consequently, $\dim(K(v)) \leq \deg(\text{minimal polynomial of } T) \leq \dim(V)$

Generalized eigenvector: $v \in V$ is a generalized eigenvector of T corresponding to eigenvalue λ if $(T - \lambda I)^k v = 0$ for some k .

- The smallest such k is the order of v (order 1 $\Rightarrow v$ is an ordinary eigenvector; order 2 $\Rightarrow v$ itself is not an eigenvector, but $(T - \lambda I)v$ is)
- If λ is not an eigenvalue of T , there is no corresponding generalized eigenvector

Theorem (analogous to Krylov subspaces): for $T \in L(V)$ and eigenvalue λ ,

- $\text{null}(T - \lambda I)^0 \subset \text{null}(T - \lambda I)^1 \subset \text{null}(T - \lambda I)^2 \subset \dots$ is a nested sequence of subspaces
- The sequence stops growing: there exists $r \leq \dim(V)$ such that $\text{null}(T - \lambda I)^r = \text{null}(T - \lambda I)^{r+1} = \dots$

Fact: $V = \text{null}(T^n) \oplus \text{range}(T^n)$, where $n = \dim(V)$ (by rank-nullity applied to T^n , and showing the intersection is $\{0\}$)

Theorem: for $T \in L(V)$ with V over $\mathbb{C}$, V always has a basis consisting entirely of generalized eigenvectors of T .

Nilpotent operator: $T \in L(V)$ is nilpotent if $T^k = 0$ for some k .

- T nilpotent $\Rightarrow 0$ is the only eigenvalue of T
- Over $\mathbb{C}$, the converse also holds: 0 is the only eigenvalue $\Leftrightarrow T$ is nilpotent

Generalized eigenspace $G(\lambda, T)$: for $T \in L(V)$ and eigenvalue λ , $G(\lambda, T) := \{v \in V : v \in \text{null}((T - \lambda I)^k) \text{ for some positive integer } k\}$

- Fact: $G(\lambda, T) = \text{null}((T - \lambda I)^{\dim V})$ (shown by proving containment both ways)

Theorem (V over $\mathbb{C}$, T has m distinct eigenvalues $\lambda_1, \dots, \lambda_m$): the following all hold (not equivalent to each other, just all true):

- $G(\lambda_k, T)$ is T -invariant for each k
- $(T - \lambda_k I)$ restricted to $G(\lambda_k, T)$ is nilpotent
- $V = G(\lambda_1, T) \oplus \dots \oplus G(\lambda_m, T)$
- Proof idea: uses that $(T - \lambda_i I)^k$ and T commute

Geometric multiplicity $g_i = \dim(E(\lambda_i, T))$ – dimension of the ordinary eigenspace

Algebraic multiplicity $a_i = \dim(G(\lambda_i, T))$ – dimension of the generalized eigenspace

Characteristic polynomial: if T has eigenvalues $\lambda_1, \dots, \lambda_m$ with algebraic multiplicities $a_1, \dots, a_m$, then $q(x) = (x - \lambda_1)^{a_1} (x - \lambda_2)^{a_2} \dots (x - \lambda_m)^{a_m}$

- Fact: $q(x) = \det(M(T, \text{basis}) - xI)$ for any basis (often easier computationally)

Trace = sum of eigenvalues (with algebraic multiplicity) = sum of diagonal entries of any matrix of T , in any basis.

Determinant = product of eigenvalues (with algebraic multiplicity). $\det(T) \neq 0 \Leftrightarrow T$ invertible.

- Both trace and det are basis-independent (similar matrices have the same trace and det)

Cayley-Hamilton Theorem: let V be finite-dimensional over $\mathbb{C}$, $T \in L(V)$, and $q(x)$ the characteristic polynomial of T . Then $q(T) = 0$.

Block diagonal matrix: a matrix of the form with square blocks $A_1, \dots, A_m$ along the main diagonal and all off-diagonal blocks equal to zero.

Theorem: let T have distinct eigenvalues $\lambda_1, \dots, \lambda_m$ with algebraic multiplicities $a_1, \dots, a_m$. There is a basis L such that $M(T, L)$ is block diagonal, with:

- number of blocks = number of distinct eigenvalues
- block A_k has size $a_k \times a_k$, with λ_k on the diagonal, and is upper triangular